Bangladesh University of Business and Technology

Course Title: Computer Networks Lab
Course Code: CSE 320

LAB REPORT

Experiment Name: VLAN Configuration

Experiment No.: **03**

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Introduction

A Virtual Local Area Network (**VLAN**) is a logical segmentation of a Layer 2 (**Data Link Layer**) network that groups devices into separate logical networks independent of their physical location. Unlike traditional LANs that depend on physical topology, VLANs are implemented on switches using **IEEE 802.1Q frame tagging**.

VLAN 2 to VLAN 1001 is considered the Normal VLAN Range. VLANs in this range are fully configurable, editable, and deletable, and are stored in the switch's VLAN database. There are two types of VLAN Links that are commonly used:

<u>**Access Link**</u>: Connects a VLAN-unaware end device (such as a PC) to a VLAN-aware switch. Frames transmitted over an access link are untagged and belong to a single VLAN.

<u>**Trunk Link**</u>: Connects VLAN-aware devices such as switch-to-switch or switch-to-router links. It carries traffic from multiple VLANs using IEEE 802.1Q tagging.

There is also a **Hybrid Link** that supports both tagged and untagged frames on the same link. It is typically vendor-specific and not part of the IEEE 802.1Q standard.

Devices and Tools

- Cisco Packet Tracer v9.0.0
- 2960-24TT Switch × 3
- Client PC × 14

VLAN Table

VLAN Number	VLAN Name
10	IT
20	HR
30	FIN

Addressing Table

Device	IP Address	Subnet Mask	VLAN Number	PORT Number
PC0	192.168.10.1	255.255.255.0	10	Fa0/1
PC1	192.168.10.2	255.255.255.0	10	Fa0/2
PC2	192.168.20.1	255.255.255.0	20	Fa0/3
PC3	192.168.20.2	255.255.255.0	20	Fa0/4
PC4	192.168.30.1	255.255.255.0	30	Fa0/5
PC5	192.168.30.2	255.255.255.0	30	Fa0/6
PC6	192.168.10.3	255.255.255.0	10	Fa0/1
PC7	192.168.10.4	255.255.255.0	10	Fa0/2
PC8	192.168.20.3	255.255.255.0	20	Fa0/4
PC9	192.168.20.4	255.255.255.0	20	Fa0/5
PC10	192.168.30.3	255.255.255.0	30	Fa0/6
PC11	192.168.30.4	255.255.255.0	30	Fa0/7
PC12	192.168.10.5	255.255.255.0	10	Fa0/1
PC13	192.168.10.6	255.255.255.0	10	Fa0/2

Procedure

The following steps were meticulously followed to setup the previously mentioned VLANs:

Preparing the Network:

- In the Cisco Packet Tracer application, we add three new **Switches (2960-24TT).** Then we add 14 Client PCs (PC0-PC13). We **interconnect** the switches.
- First two **Switches** are connected to six **client PCs** and the last **Switch** is connected to two **client PCs.** We connect the network devices using the **Auto Cabling** tool in the Cisco Packet Tracer application.

Configuring the client PCs:

- Single click the **PC0** and a new window appears.
- Under the **Desktop** tab, go to IP Configuration.
- Add the **IPv4 Address** according to the aforementioned **Addressing Table**. Repeat the steps for **PC1-PC13**.

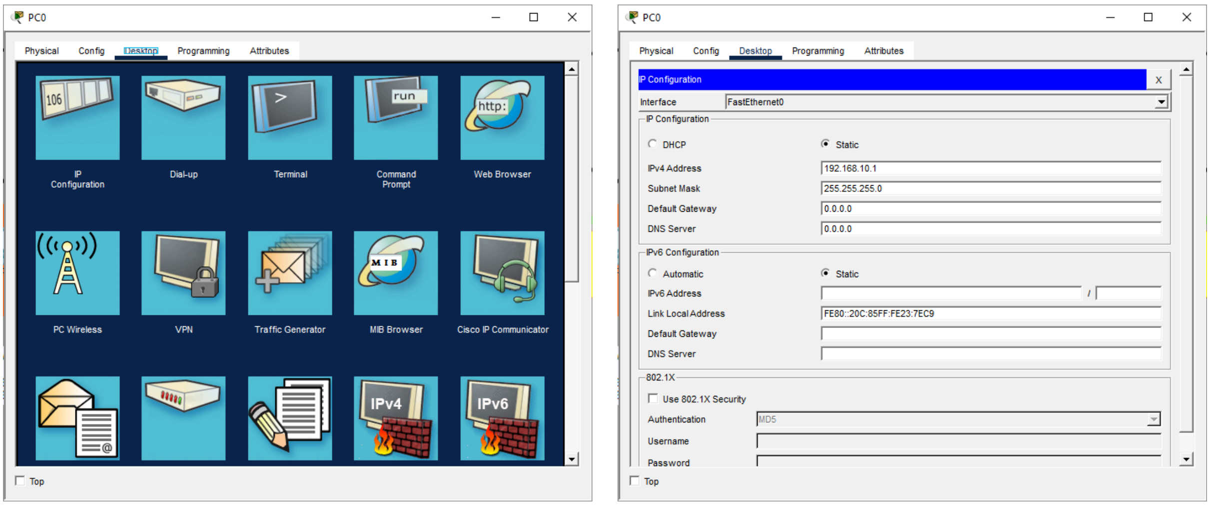

Fig 1.1: Configuring the client PCs.

Configuring the Switches:

- Click **Switch0** and a new window will appear.
- Under the **Config** tab, we click **VLAN Database**. We add three VLANs by their name and number:
  - IT (10)
  - HR (20)
  - FIN (30)
- We can see **Interfaces** ranging from FastEthernet0/1 to FastEthernet0/24. By using the **Addressing Table**, we assign each of the **FastEthernet** ports their respective VLAN number. We repeat this **for all the clients connected to this Switch**.
- When connecting two Switches through a port, we must configure it as **Trunk**.
- We repeat all the steps for **Switch1** and **Switch2**.

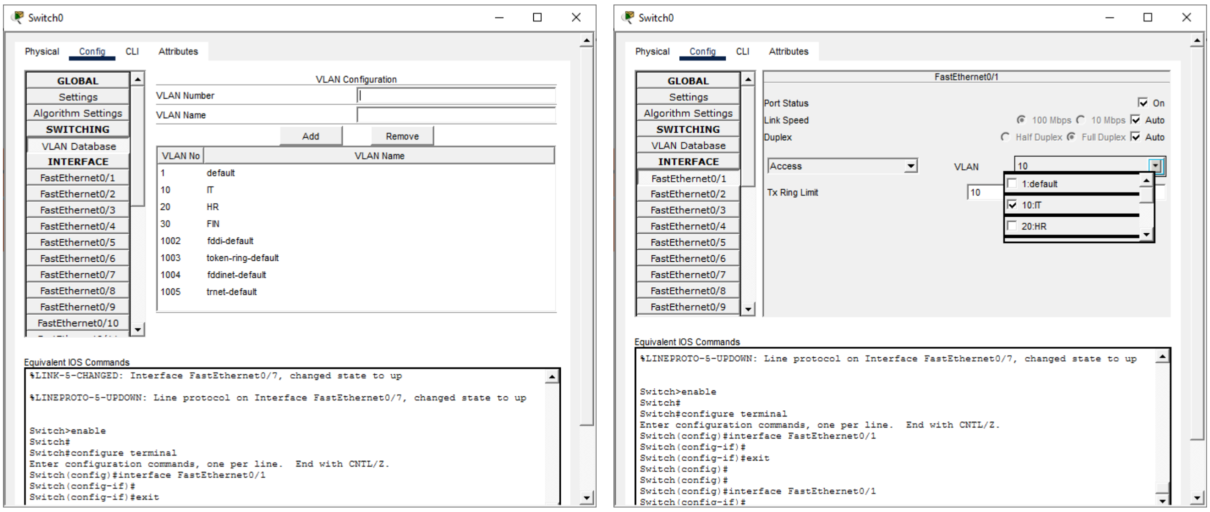

Fig 1.2: Configuring the Switches.

Isolating the Virtual Local Area Networks:

- It is possible to isolate the client devices directly in the Network diagram.
- From the toolbar, Click the **Ellipse tool** or **type "E"** in the keyboard for shortcut. A new palette dialog will appear. Uncheck **Outline** and check **Fill Color**.
- Select any appropriate fill color and draw ellipses around the client PCs.
- This step is optional however, it adds the much needed visual separation for the VLANs.

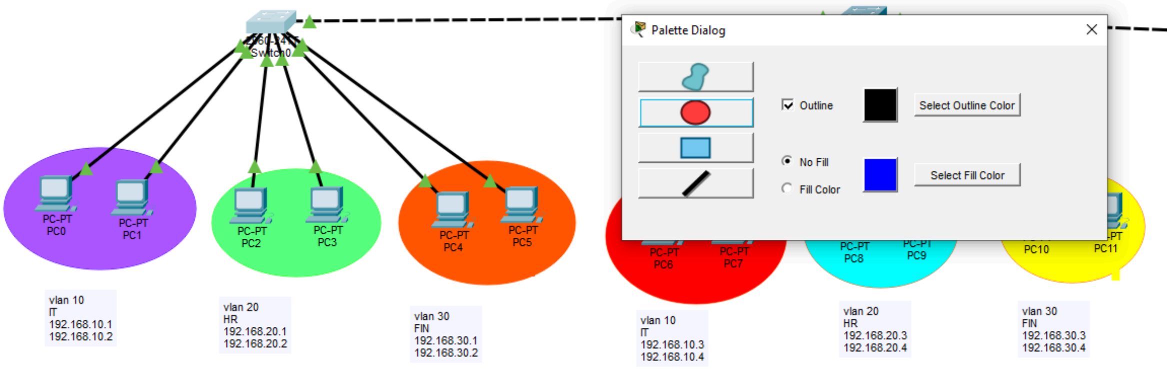

Fig 1.3: <u>Isolating the VLANs.</u>

Configuring VLAN using CLI

It is possible to configure the VLANs using the command line interface instead of doing it manually. Before doing that, we must prepare the network and connect the devices as described in the procedure.

Configuring the Switches:

- Click **Switch0**, a new window appears. Click the **CLI** tab.
- Type **ENTER** until you see **“Switch>”**
- Type the following command to enter the Switch to **configuration mode:**

```
Switch> enable
Switch> config t
```

Add to VLAN Database:

The following lines of code are executed for **Switch1** and **Switch2** as well:

```
Switch> vlan 10
Switch> name IT
Switch> exit
Switch> vlan 20
Switch> name HR
Switch> exit
Switch> vlan 30
Switch> name FIN
Switch> exit
Switch> do sh vlan
```

Add VLAN information to client ports:

The following lines of code are executed for **Switch1** and **Switch2** as well:

```
Switch> int range Fa0/1-2
Switch> switchport mode access
Switch> switchport access vlan 10
Switch> exit
```

```
Switch> int range Fa0/3-4
Switch> switchport mode access
Switch> switchport access vlan 20
Switch> exit
Switch> int range Fa0/5-6
Switch> switchport mode access
Switch> switchport access vlan 30
Switch> exit
Switch> do wr
```

Connect two Switches:

We have to identify which port(s) connect any two switches and use the port number to specify it as **“Trunk”**:

```
Switch> int range Fa0/7
Switch> switchport mode trunk
```

Result

Once we have followed all the steps mentioned in the previous sections, we should have a working VLAN configuration of three VLANs (IT, HR, FIN) and 14 client PCs:

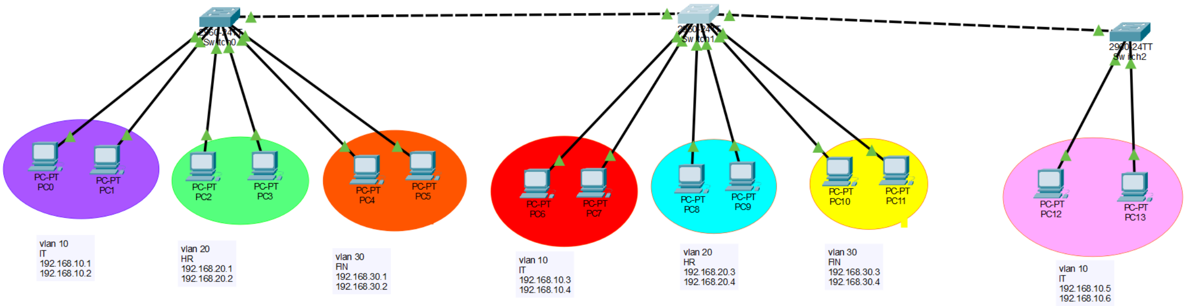

Fig 2.1: <u>VLAN Configuration.</u>

Testing and Verification

We have to test whether client PCs can communicate under the same VLAN and whether the communication fails within different VLANs:

Testing communication under same VLAN:

- From the toolbar, click **"Add Simple PDU"**. The cursor would turn into a message icon.
- Click on **PC0** and then under the same VLAN, click **PC13**.
- If everything goes as expected, we should see a successful communication.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC13	ICMP		0.000	N	0	(e...	

Fig 3.1: <u>Testing communication under the same VLAN.</u>

Testing communication under different VLANs:

- From the toolbar, click **"Add Simple PDU"**. The cursor would turn into a message icon.
- Click on **PC1** and then under the same VLAN, click **PC3**.
- If our VLAN configuration is OK, we should see a failed communication.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC13	ICMP		0.000	N	0	(e...	
	Failed	PC1	PC3	ICMP		0.000	N	1	(e...	

Fig 3.2: Testing communication under different VLANs.

Conclusion

The objective of the third experiment was successfully achieved by configuring three different VLANs using 14 client PCs. It provided core skills for network isolation and enhanced data privacy.